

1 Introduction

This sprint focused heavily on stabilizing and refining our self-driving car system after resolving several hardware and calibration issues. The replacement of a broken servo arm led to a cascade of changes, requiring a full recalibration of both the steering and speed systems. These adjustments impacted core components like our steering regulators, prompting updates across the entire codebase. While most issues have been resolved, some challenges such as lane following on dotted lines remain and are scheduled for the next sprint. Alongside these fixes, we implemented major new features like ramp navigation, roundabout handling with camera-based exit detection, and smarter intersection alignment. We've also made key decisions regarding hardware and introduced new track elements to support path planning.

2 Planned activities

During this sprint, we had a plan to accomplish the following tasks:

- Optimize current codebase wherever possible
- Mechanical maintenance
- Finish roundabout handling and overtaking
- Integrate localization for path planning/replanning

3 Status of planned activities

- **Optimize current codebase wherever possible**

This task proved to be bigger than we anticipated. As mentioned, physical changes prompted a lot of fixes in the codebase. As for things that didn't break first before they were optimized, we managed to triple the speed of the pre-processing adjustments for our camera frames (30ms to 10ms).

- **Mechanical maintenance**

As mentioned in the previous report, at the end of the last sprint our steering servos arm finally gave out and broke. We had trouble finding a good replacement, and the best one we could find was a few millimeters shorter than the original. This caused us to completely take apart the steering system and put it back together in a way that no parts were touching each other. When that was done, we dropped the 'zip ties and tape' model and fixed our microcontroller and sensors to the shell of the car, so that we wouldn't need to worry about disconnecting and reconnecting wires when putting the shell on or taking it off.

- **Finish roundabout handling and overtaking**

As of right now, the car can handle roundabouts fairly well. That being said, parts of the way it handles roundabouts are still pre-written sequences which is why it sometimes goes over the road markings. We have thought of a way to fix this, and we are confident that we will implement the fix during the next sprint. As for overtaking, we have implemented it code-wise, however, we still haven't figured out a way to test it as our track is fairly small and we don't have adequate space.

- **Integrate localization for path planning/replanning**

Our initial idea for this was to place a rotary encoder on the car for basic odometry. After some consideration, we realized that for this approach, we would need more encoders than we can fit on the car, so we gave up on it. We have more ideas in the works, and we intend to implement one of them for the next sprint, as soon as we decide on the best approach.

4 General status of the project

During this sprint, we fixed lots of issues. After we replaced our broken servo arm, we noticed that our speed wasn't calibrated. Naturally, we calibrated both the speeding and steering systems, but unfortunately that caused issues with the rest of our code, mainly the regulators we made for steering. Since two of our most important systems underwent huge changes, we now needed to adjust almost everything else to accommodate those new changes. Most notably, the lane following algorithm starts veering left to right when the road markings are dotted. We have fixed pretty much everything else besides that, and due to Easter closing in, we decided to leave that for the next sprint. Our most notable new features are driving over ramps, taking roundabouts, detecting stationary cars and roundabout exits with the camera, and a new algorithm for the car to align itself with the road properly if it enters an intersection at a weird angle. Also, we've decided not to use a polarizing film as it was interfering with object detection and the risk to reward ratio of yet another dataset was too high for our liking. As for the track, we've added a starting area, a ramp and a roundabout, and mapped them out for our path planning algorithm.

5 Upcoming activities

In the next sprint we want to do the following tasks:

- Make sure everything works with new calibrations, remove bugs
- Fix dotted line following
- Implement localization
- Keep optimizing everything